

SHRISTI KESHRI

SOFTWARE AND AI ENGINEER

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Software & AI Engineer with a Computer Science degree from The Ohio State University (AI specialization, Mathematics minor) and hands-on experience building retrieval-augmented generation (RAG) pipelines, tool-calling AI agents, and full-stack web applications. Specialized in hybrid retrieval systems, LLM evaluation, and agentic AI workflows, currently applied as an AI Intern at Matellio Inc. Proven track record of shipping working systems end-to-end — from model and retrieval logic to API and interface — across internships, hackathons, and independent projects.

TECHNICAL SKILLS

Software Design & Architecture: API design, component architecture, system design for scalable services, request/response schema validation

Programming & Development: Python, Java, C, C++, JavaScript, TypeScript, SQL

AI / Machine Learning: LangChain, Groq API, hybrid retrieval (BM25 + FAISS), cross-encoder reranking, LLM evaluation, PEFT/LoRA fine-tuning, YOLOv8, OpenCV, MediaPipe, Scikit-learn

Data & Database Systems: NumPy, Pandas, ChromaDB, Supabase

Web & Backend: React, React Router, Axios, FastAPI, Flask, Pydantic, Node.js, REST APIs

DevOps & Tools: Git/GitHub, Docker, Google Colab, Hugging Face Spaces, VS Code

PROFESSIONAL EXPERIENCE

AI Intern, Data Science & AI

May 2026 – Present

Matellio Inc., Kolkata, IN (Remote)

- Engineered a hybrid retrieval pipeline (BM25 + FAISS + cross-encoder reranking) over a 500-document corpus, achieving Precision@3 of 0.80 and MRR of 0.82 across 25 evaluation queries
- Improved Precision@3 from 0.40 to 0.467 on a smaller test set by adding cross-encoder reranking to the retrieval pipeline
- Built tool-calling, memory-equipped AI agents using LangChain and the Groq API, deployed behind a FastAPI inference server with Pydantic-validated request/response schemas
- Hardened the inference API against malformed requests and schema errors by authoring a 16-test validation suite, reducing the risk of faulty inputs reaching the LLM pipeline in production
- Benchmarked Llama 3.1-8B, Llama 3.3-70B, and Phi-3 across latency, output quality, and cost, identifying inference bottlenecks via local Ollama deployment
- Tuned document chunking strategies (size and overlap) to improve downstream RAG answer quality across the evaluation set

Web Development Intern

May 2024 – Nov 2024

PATH Pod, Positive Approaches to Health Podcast, Columbus, OH

- Owned site performance and reliability end-to-end as the primary engineer on the project
- Improved frontend performance by 40% through asset compression, lazy loading, and JavaScript optimization
- Resolved cross-browser compatibility issues and shipped responsive, adaptive UI using Git and Chrome DevTools

Undergraduate Teaching Assistant

May 2023 – Aug 2023

OSU Dept. of Computer Science & Engineering, Columbus, OH

- Mentored 50+ students through core Java programming concepts during weekly lab sessions, providing hands-on, one-on-one debugging guidance
- Delivered detailed, actionable feedback on coding assignments, identifying recurring misconceptions and helping students build stronger debugging habits rather than just fixing the immediate error

Student Manager (prev. Student Lead, Student Barista)

Aug 2021 – May 2025

Campus Grinds, OSU Dining, Columbus, OH

- Promoted twice over 4 years — from Barista to Student Lead to Student Manager — based on consistent performance and leadership shown on the floor

- Built and led teams across 10 high-volume campus cafés, training and onboarding 200+ student employees on service standards, food safety, and conflict resolution
- Owned end-to-end café operations, coordinating schedules, delegating tasks, and managing inventory (FIFO) and food donation logistics to local shelters while balancing a full course load
- Awarded Student Manager of the Year 2024 for outstanding leadership, operational excellence, and team collaboration across the dining program

PROJECTS

BioKey — Iris-Based Biometric Authentication System

Jun 2026 – Present

- Engineered a real-time iris recognition pipeline using OpenCV and MediaPipe's FaceLandmarker model for landmark-based feature extraction, with an enrollment-and-verification flow built on similarity-threshold biometric matching
- Optimized the inference pipeline for edge deployment, achieving ~15ms per-scan processing time on CPU-only hardware (M1 MacBook Air) — fast enough for real-time, on-device verification without cloud dependency
- Productionized the end-to-end CV pipeline as a Flask REST API (368 lines across 6 files) and deployed it to Hugging Face Spaces for public access

HireHub — AI Resume Analyzer

Mar 2026 – Present

- Owned the frontend integration for an AI-powered resume analyzer, calling the resume-analysis API via Axios and designing the workspace UI to surface AI-generated analysis results and feedback to users
- Implemented authentication (register/login/session, protected routes) and resume CRUD integrated with a FastAPI backend, as frontend owner on a 2-person team
- Designed component architecture in React using React Router with a consistent design system across landing, dashboard, and workspace pages; shipped a full, navigable multi-page app with end-to-end auth flow (currently paused with core architecture in place for resumption)

Safe Sound — 11th MAKE OHI/O Hackathon, AEP Challenge (2nd Place)

Mar 2025

- Integrated YOLOv8 image detection with audio classification into a combined, real-time multi-modal alert pipeline
- Engineered fusion logic to cross-validate signals before triggering an alert, reducing false positives by 35%
- Achieved 92% event detection accuracy with sub-150ms latency per frame, winning 2nd place out of competing teams

EDUCATION

B.S. Computer Science & Engineering

Graduated May 2025

The Ohio State University, College of Engineering, Columbus, OH — Specialization: Artificial Intelligence; Minor: Mathematics

Relevant Coursework: Data Mining, Machine Learning, Introduction to Artificial Intelligence, Statistics, Algorithms, Data Structures

HONORS & AWARDS

- Judges' Choice Award, ASA DataFest
- 2nd Place, Safe Sound, 11th MAKE OHI/O Hackathon (AEP Challenge)
- 2nd Place, Park Assist, 10th MAKE OHI/O Hackathon (Honda Challenge)
- Student Manager of the Year 2024, OSU Dining Services

CERTIFICATIONS

- Code Generation and Code Optimization using IBM Granite, IBM SkillsBuild — Issued Nov 2025
- Getting Started with Python, University of Michigan, Coursera — Issued Jun 2021